

When Is Unsurprising Science Informative?

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Imagine a scientific study about to be conducted. Before it is conducted, you ask a group of scientists what they expect it to find. Everyone gives you the same answer: X. The study is conducted, and X is exactly what the study finds. Whatever else one might say about the result, it is not surprising. The evidence accords with what scientists already expected to see.

But—how much have they learned?

That depends on something the absence of surprise does *not* tell us. It depends on *why*, exactly, scientists expected that result in the first place. Or, in other words, it depends on what it means when we say the result was “what scientists expected.” Because not all expectations are created equal.

Imagine that in one case scientists expected X because the relationship under study had survived decades of rigorous testing across different methods and settings. In another, they expected X because an influential theory predicts it, a few suggestive studies seemed consistent with it, and over time the claim has become conventional wisdom. The same result, X, would be equally unsurprising relative to the stated expectation in both cases. Yet, another confirmation of the first claim can be thought to add very little to scientific knowledge, while careful evidence in the case of the second could add a great deal.

If everyone expects X, showing X may be unsurprising. It does not follow that it is uninformative. That depends on why everyone expects X.

To unpack this idea, we have to be clear about one thing up front: none of this is to say that surprise lacks value. Scientists have good reasons to value unexpected results. A result that violates a well-established expectation can expose a weakness in existing theory or reveal an assumption that has gone unnoticed. In Kuhn’s account, persistent anomalies of this kind can contribute to periods of crisis and, sometimes, to the larger

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reorientations of scientific practice—his famous “paradigm shifts” (Kuhn 1971). Surprise can therefore matter because of what it forces science to reconsider.

It also matters socially. Murray Davis famously argued that ideas become *interesting* when they deny some part of an audience’s taken-for-granted “assumption-ground” (Davis 1971). More recently, James Evans, Casey Petroff, and Gary King have pushed this intuition further, arguing in *Science* that scientists should become more deliberate about choosing questions, audiences, and framings that maximize surprise, because findings that move an audience farther from its expectations can contribute more to collective knowledge.

There is an important insight here. But that insight raises a key question: What gives an expectation the epistemic standing that makes its violation informative?

Surprise measures a relationship between an expectation and an observation. It alone does not tell us where that expectation came from in the first place and, crucially, whether it is well founded.

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At first blush, scientific expectations might seem to have an obvious source. Scientists immerse themselves in the literature. They accumulate knowledge of the literature’s findings, they evaluate the quality of different research designs, and they learn which results have survived scrutiny across methods and settings. On this idealized account, what a scientist expects is something like a carefully considered summary of the available scientific evidence. And often it is. To a point.

But most scientists will also recognize a kind of conversation that illustrates a complication to that account. Someone proposes a study and a colleague asks, *Don’t we already know that?* Sometimes the answer is plainly yes. There is a deep, rigorous, durable body of evidence behind the expected result.

But sometimes, when pressed on exactly *how* we supposedly know, the answer becomes less definite. Well... the result fits a familiar, but poorly tested—even wholly untested—theory. Or: it seems consistent with what we remember from the literature. Maybe it accords with personal experience. Or it simply *makes sense*. What initially sounded like a statement about established scientific knowledge turns out to be, in meaningful part, a judgement about something much more like *plausibility*.

This doesn’t happen because scientists are careless. Instead, it’s because science is hard, and hard in ways *other* than those we typically take stock of. Scientific expectations cannot simply be read off from the literature. Scientists have to *decide* which evidence bears on the question at hand, which findings deserve greater weight,

and what a result obtained in one setting implies for another. Theory enters into those judgments, along with methodological commitments, and disciplinary conventions.

But so too is that decision-making subject to tacit background knowledge, intuitions about what is plausible, and broader cultural values about what a satisfactory explanation should look like. Physicists, for example, have long appealed to simplicity, symmetry, and mathematical elegance in judging which theories are promising. As Poincaré described aesthetic sensibility, it constitutes a “delicate sieve” through which scientists form expectations and select research possibilities worth pursuing. The evidence matters enormously, yes—but to claim that it is the *only* basis on which expectations are formed denies a large body of evidence on the sociological and cultural practice of science.

Furthermore, the judgement that scientists exercise takes place under constraint. No scientist can continuously reconstruct the evidentiary pedigree of every claim they rely upon. Years of training and accumulated expertise help enormously, but experts still have to compress information; they remember some findings more readily than others; and they often come to rely on what a field has already come to treat as settled.

Nor is the scientific record they encounter neutral. We know from a large literature on the economic organization of science that publication, citation, visibility, prestige, timeliness, and enthusiasm all sort findings according to criteria that do not always track their evidentiary strength. Some results travel farther and faster than others. A striking finding can become part of a field’s common knowledge even while its durability remains uncertain. A quieter but better-supported result, by contrast, can attract less attention.

Scientists therefore form expectations from a body of knowledge whose contents and salience have already been shaped by institutions that do not necessarily privilege the most securely established claims.

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Once expectations are understood this way, a crucial distinction appears between what a scientific community *accepts* and how strongly that acceptance is *warranted*. Often the two move together. Claims supported by extensive and reliable evidence become accepted for good reason. But, importantly, the two factors do not necessarily move together in practice. A proposition can become accepted—at least to the point that it meaningfully shapes expectations—before its evidentiary foundation is secure.

This is why Davis's "assumption-ground" is not a simple epistemic ledger. It captures what a given audience has come to take for granted. But, it alone does not tell us how what is taken for granted got there, or how much warrant presently stands behind it.

Once the response to the claim of X becomes "*of course*," that response in and of itself cannot tell us whether X rests on a strong warrant or not.

Philip Kitcher's account of scientific knowledge gives a particularly clear articulation of this problem. Cumulative science, in his formulation, requires an inherited store of claims that researchers can rely upon without reconstructing the entire evidentiary history of each one. At some point, claims have to go "on the books" so that later inquiry can build from them rather than begin again. That is indispensable to scientific progress. But it also means that better individual vigilance can never fully close the gap between acceptance and warrant—a claim's status as accepted knowledge can travel through a scientific community much more easily than the detailed warrant that originally supported it.

There are, of course, ways that inherited knowledge can be checked again without scientists reconstructing its entire evidentiary history. One is through use. When accepted knowledge is repeatedly *used* to make correct predictions, to build things, or to successfully intervene in the world, its performance supplies an additional check on whether those claims deserve their standing.

But where such feedback has limited opportunity to occur, is weaker, slower, or difficult to interpret, then accepted claims may circulate for longer without that form of independent test. This is not to say that practically useful science is necessarily better science. It also doesn't mean that fields without immediate applications produce weaker knowledge. It is simply to note that repeated successful *use* of knowledge can provide one more way for the warrant behind an accepted claim to be tested and substantiated, not just inherited.

Scientific inquiry, meanwhile, does not wait for history to finish sorting the durable findings from the transient ones. Today's accepted knowledge becomes part of tomorrow's expectations while that sorting is still underway. What this means is that what a field takes for granted can get ahead of what it has *earned the right* to take for granted.

Once we recognize acceptance and warrant can become untethered, surprise alone cannot tell us how much a new finding teaches us. It tells us how far an observation departs from what scientists expected. It does not tell us how much evidentiary work remained to be done on the expectation itself.

Importantly, the consequence runs in both directions. Evidence that agrees with an expectation can leave our best answer exactly where it was while giving us substantially better warrant to believe it. Thus: an unchanged *answer* is not the same thing as unchanged *knowledge*.

While on the other hand, a study expected to confirm what everybody *thinks* they know may do something else entirely: show that everybody was wrong. We cannot always know in advance which questions will surprise us with their answers.

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So, *when is unsurprising science informative?* When an expectation is more settled than the evidence behind it warrants. It is informative when additional evidence can materially strengthen our *reasons* for believing the expected answer—and potentially more informative still when the expectation itself turns out to be wrong.

This matters for more than how completed research is evaluated. It also matters for what scientists choose to study in the first place. Judgments about novelty and prospective surprise inevitably help allocate limited attention. Evans, Petroff and King, for instance, urge scientists to ask questions in anticipation of surprise.

When anticipated agreement with existing expectations becomes a reason *not* to investigate, however, the possibilities for scientific knowledge suffer. When that happens, surprise begins to shape not only how evidence is received, but which evidence will ever exist. Studies expected to challenge prevailing understanding are more likely to be undertaken; those expected merely to reinforce it are easier to dismiss. The missing evidence is not sitting in the epistemic ledger, merely unseen. Instead, the experiment is never run at all.

“We already know the answer”—that familiar refrain to a proposed study—can therefore mean two very different things. Sometimes it is a justified stopping rule. We have enough reliable evidence that another study won’t teach us much. Sometimes, though, it means only that a scientific community has a strong expectation. And as we’ve argued, strong expectations are not necessarily warranted expectations.

Of course, science must eventually stop investigating some things. A cumulative enterprise could not function otherwise. The question is how to know the time has come. The fact that an answer would no longer surprise us is not, by itself, enough.

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Surprise deserves its prominent place in science. When well-founded expectations fail, we should pay attention. But science also progresses when the world behaves as expected and we acquire better reasons for believing that it does—and sometimes by

investigating what everybody expected, only to discover that the world does something else entirely.

The distinction worth preserving is not between surprising and boring science. It is between what we have come to expect, and what we have earned the right to treat as known.